



BASIC MOLECULAR **BIOLOGY TECHNIQUES**

1 Month Training Report

BY

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BTECH BIOTECHNOLOGY

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JACOB INSTITUTE OF BIOTECHNOLOGY AND BIOENGINEERING

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ACKNOWLEDGMENT

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INTRODUCTION

About Institute

The Indian Institute of Sugarcane Research (IISR), Lucknow was established in 1952 by the erstwhile Indian Central Sugarcane Committee for conducting researches on fundamental and applied aspects of sugarcane culture as well as to co-ordinate the research work done on this crop in different States of the country. The Government of India took over the Institute from the Indian Central Sugarcane Committee on 1st January, 1954. On April 1, 1969, it was transferred to the Indian Council of Agricultural Research (ICAR), New Delhi along with other central agricultural research institutes.



Location

The Institute is located 4.8 km from Lucknow Railway Station (Charbagh), 8 km from Alambagh Bus Station, and 12 km from Chaudhary Charan Singh International Airport Lucknow at Rae Bareilly Road (26.56°N and 80.52°E) at 111 m above mean sea level. Distance from New Delhi is around 500 km and well connected to all main cities of the country.

Climate

The climate of the location is subtropical semi-arid. The summer season, spanning from mid-March to mid-June, is hot and dry with mercury exceeding 40°C quite frequently in recent past. The long term monthly average maximum temperature for April, May and June are 36.4, 40.0 and 38.5°C. The monsoon normally commences from 18th-21st June and extends till last week of September. The average rainfall during last five years has been around 880 mm and rainy season is primarily unimodal (nearly 80-85 % of precipitation received during monsoon). The winter season starts from mid-November and is extremely chill with mercury touching about 1°C at times. The long-term average minimum temperature for November, December, January and February are 11.5, 7.3, 7.0 and 9.7°C.

Division and Sections:

There are many departments in ICAR-IISR, Lucknow.

1. Crop Improvement
2. Crop Protection
3. Crop Protection
4. Plant Physiology & Biochemistry
5. Agricultural Engineering
6. Extension & Training
7. Agricultural knowledge Management Unit
8. Project Monitoring & Evaluation

9. Agro-meteorology

10. Jaggery Unit

Division of Crop Improvement

The Division of Crop Improvement owes its genesis to the establishment of Botany and Breeding unit at ICAR-Indian Institute of Sugarcane Research, Lucknow by Sugarcane Breeding Institute, and Coimbatore for evaluation of sugarcane varieties in sub-tropical region. This unit was transferred to IISR, Lucknow in 1969 and a section of Botany and Breeding came into existence. This section was entrusted with the responsibility of genetic evaluation of sugarcane crosses in subtropical India and germplasm evaluation of sugarcane and sugar beet. Later on, this section was elevated to the Division of Botany and Breeding. It was renamed as Division of Crop Improvement in 1989. As per Institute's mandate, this Division is conducting fundamental and applied research on sugar crops (sugarcane and sugar beet). Evaluation and enrichment of germplasm was also initiated for its utilization for sugarcane varietal improvement programs in subtropics.

Mandate

- Development of superior sugarcane varieties with special reference to subtropics.
- Evaluation and documentation of sugarcane germplasm for North Indian conditions.
- Development of breeding and genetic stocks for specific attributes.
- Application of biotechnological approaches for sugarcane improvement.

Mission

- To develop sugarcane varieties resistant to red rot for higher sugar productivity.

Thrust Areas

- To develop suitable varieties for sub-tropical India
- Genetic enhancement in traits of economic importance through gene introgression from *Saccharum* complex
- Increasing breeding efficiency through biotechnological interventions
- Sugar beet breeding for Indian agro-climates and its adaptation

FIELD VISIT
DIVISION OF CROP IMPROVEMENT



Figure 1 : Division of Crop Improvement Mist Chamber , ICAR-IISR, Lucknow

DIVISION OF CROP PRODUCTION

Figure 2 : Division of Crop Production , ICAR-IISR, Lucknow

MORPHOLOGICAL DATA ANALYSIS

MATERIAL

12 Varieties of sugarcane were planted in the early spring season of 2021 in the research field of ICAR - Indian Institute of Sugarcane Research Lucknow. All the recommended agronomic and plant protection measures were adopted for raising a healthy crop. Morphological data analysis was done with the 12 varieties of sugarcane in the month of December 2021.

METHOD

The sugarcane varieties were washed properly, dried and then the data was recorded. Data was recorded on measurable and morphological characters at early stage of crop growth and the grand growth stage of the crop. The data was collected in the grand growth stage for its comparison with the early growth traits. For every variety, two clumps were chosen and the data was collected. To obtain a mid-value between the two clumps, the mean was calculated of the two clumps of one variety for every trait.

Characters to be determined-

LEAF:

- No. of green leaves
- Leaf length(cm)
- Leaf width (cm)
- Leaf area (cm²)
- Leaf fresh weight (gm)
- Leaf dry weight (gm)
- Moisture content %

STALK:

- No. of tillers
- Stalk length (cm)
- Number of nodes
- Length of internodes (cm)
- Stalk fresh weight (cm)
- Stalk dry weight (cm)
- Moisture content %

ROOT:

- Root length (cm)
- Density (gm/cm³)
- Fresh weight (gm)
- Dry weight (gm)
- Moisture content %



Figure 3 : Morphological Data Analysis of Sugarcane samples in the month of December, 2021



Figure 4 : Morphological Data Analysis of Sugarcane samples in December, 2021



Figure 5 : Morphological Data Analysis of Sugarcane samples in December, 2021

Leaf traits

- **Number of green leaves** - The total number of green leaves present in a clump.
- **Leaf length (cm)** - Leaf blade length of the longest furled leaf (+6 leaves from spindle leaves).
- **Leaf width (cm)** - Leaf width at the broadest point was measured in cm on the same plants.
- **Leaf area (cm²)** - Leaf area was also calculated.
- **Leaf fresh weight (gm)** - All the leaves of a clump were separated from the rest of the plant. The leaves were tied together and weighed using a weighing machine.
- **Leaf dry weight (gm)** - The leaves were dried in a hot air oven and measured using a weighing machine for each clump.
- **Leaf moisture content (%)** - Moisture content of the leaves was calculated using the formula:

$$MC = \frac{w - d}{w} \times 100$$

w = wet weight

d = weight after drying

Table 1 : Leaf Traits for Morphological Data Analysis of Sugarcane Samples (December 2021)

Leaf Traits (21-12-2021)								
Sl. No.		No. of green leaves	Leaf length (cm)	Leaf width (cm)	Leaf area (cm ²)	Fresh weight (gm)	Dry weight (gm)	% Moisture content
1.	Clump 1	38	108.8	3.7	22.98	234	112	186.1367521
	Clump 2	16	91	3.2	23.36	88.5	37.5	46.12711864
2.	Clump 1	36	102	4.5	29.93	192	161.5	107.8854167
	Clump 2	24	66	3	16.70	95.0	35.5	57.63157895
3.	Clump 1	40	97	3.2	17.40	175.0	123	104.7142857
	Clump 2	25	100	3.6	21.76	135.5	48	100.0756458
4.	Clump 1	23	109.5	3.5	24.84	177	129	104.1186441
	Clump 2	45	94	2.3	15.64	197.5	73	160.5379747
5.	Clump 1	41	115	4	29.68	189.5	117	127.7585752
	Clump 2	16	100	4.2	31.23	81.0	29.5	44.58024691
6.	Clump 1	43	145	4.2	11.50	441.5	355.5	360.9790487
	Clump 2	19	126	4	39.04	125	52	83.4
7.	Clump 1	60	119	3	26.49	386	228.5	326.8031088
	Clump 2	31	112	3.7	30.87	167.5	83	117.9477612
8.	Clump 1	38	113.5	3.8	16.05	255.5	211.5	172.7211135
	Clump 2	36	155	4.1	44.15	240.5	139.5	182.495842
9.	Clump 1	54	100	3.5	11.48	290.5	139.5	242.479346
	Clump 2	23	100	2.2	14.49	75.0	31	33.66666667
10.	Clump 1	38	104	3.3	18.61	193.5	131	125.7997416
	Clump 2	22	126	3.3	23.19	93.5	44.5	45.90641711
11.	Clump 1	40	104.5	2.9	20.55	237.5	114.5	189.2894737
	Clump 2	32	100	3.7	23.13	135	52	96.48148148
12.	Clump 1	34	132	4.2	30.57	253.5	213.5	169.2790927
	Clump 2	22	94	3	22.14	114	69.5	53.03508772



Figure 6 : Leaf traits of Sugarcane samples

Stalk Traits-

- **No. of tillers:** Total numbers of tillers present in a clump were counted.
- **Stalk length (cm):** Cane stalk height was measured from bottom to top.
- **Number of nodes:** Sugarcane stalks are made up of several segments known as joints. At each joint is a node. The total numbers of nodes present in a stalk were counted.
- **Length of internodes (cm):** The distance between two nodes are long sections called internodes. The length of internodes was measured using a measuring scale in cm.
- **Stalk fresh weight (cm):** The stalk of a clump was separated from the rest of the plant and weighed using a weighing machine.
- **Stalk dry weight (cm):** The stalk was dried in a hot air oven and measured using a weighing machine for each clump.
- **Moisture content %:** Moisture content of the stalk was calculated using the formula:

$$MC = \frac{w - d}{w} \times 100$$

w = wet weight

d = weight after drying

Table 2 : Stalk Traits for Morphological Data Analysis of Sugarcane Samples (December 2021)

Stalk Traits (21-12-2021)									
Sl. No.		No. of tillers	Stalk length (cm)	No. of nodes	Length of internodes (cm)	Fresh weight (gm)	Dry weight (gm)	% Moisture content	Brix
1.	Clump 1	5	125	17	8	1515	1306	1428.79538	22
	Clump 2	4	133	13	12	838	671	757.928401	20
2.	Clump 1	2	136	13	8	717	631	628.9944212	18
	Clump 2	5	130	11	10.3	470	360	393.4042553	18
3.	Clump 1	5	125	11	10	1077.5	926.5	991.5139211	18
	Clump 2	6	109	11	10	736.5	459.5	674.1103191	20
4.	Clump 1	3	75	9	4	529.5	351	463.2110482	18
	Clump 2	4	108	9	10	901.5	707	823.0751525	16
5.	Clump 1	3	85	7	8.2	705	549.5	627.0567376	22
	Clump 2	3	100	12	6.6	789	531.5	721.6362484	18
6.	Clump 1	7	110	7	12.5	1618	1306.5	1537.252163	18
	Clump 2	3	133	12	11	815	507	752.791411	20
7.	Clump 1	5	120	9	12.5	2306.5	1990.5	2220.200412	18
	Clump 2	3	149	18	8.8	349.5	466.5	216.0236052	18
8.	Clump 1	4	166	22	9	1669.5	1235	1595.525756	22
	Clump 2	4	158	17	10	1526.5	1128.5	1452.572715	24
9.	Clump 1	8	100	16	8	905	794	817.2651934	20
	Clump 2	4	130	8	13.5	1196	803.5	1128.817726	20
10.	Clump 1	4	129	11	12	863	634	789.5353418	18
	Clump 2	5	120	10	9.6	1515	1149	1439.158416	20
11.	Clump 1	6	117	15	10	875	837.5	779.2857143	20
	Clump 2	4	136	14	8	1312.5	966	1238.9	20
12.	Clump 1	7	145	20	9	933.5	829.5	844.6408677	22
	Clump 2	5	130	15	8.6	759	491	694.3096179	20



Figure 7 : Red rot of Sugarcane sample



Figure 8 : Red rot of Sugarcane sample



Figure 9 : Stalk traits of Sugarcane samples

Root Traits-

- **Root length (cm):** The length of the longest root was measured
- **Density (gm/cm³):** The root was dipped in a measuring cylinder and the amount of water displaced by the root was measured.
Density = Fresh weight/ volume of water displaced
- **Fresh weight (gm):** The root of a clump was separated from the rest of the plant and weighed using a weighing machine
- **Dry weight (gm):** The root was dried in a hot air oven and measured using a weighing machine for each clump.
- **Moisture content %:** Moisture content of the root was calculated using the formula:

$$MC = \frac{w - d}{w} \times 100$$

w = wet weight

d = weight after drying

Table 3 : Root Traits for Morphological Data Analysis of Sugarcane Samples (December 2021)

Root Traits (21-12-2021)						
Sl. No.		Root length (cm)	Density (gm/cm ³)	Fresh weight (gm)	Dry weight (gm)	% Moisture content
1.	Clump 1	16	1.1375	33	20	46.15384615
	Clump 2	17	0.6	62	28.5	33.33333333
2.	Clump 1	21	1.9	15	12.5	26.31578947
	Clump 2	21	0.825	27	16.5	21.21212121
3.	Clump 1	15.5	2	33	23	30
	Clump 2	18.5	1.05	49	26	26.19047619
4.	Clump 1	8.5	3.625	19.5	14	27.5862069
	Clump 2	21.5	1.175	22.5	18.5	27.65957447
5.	Clump 1	14	1.166	17	13	17.85714286
	Clump 2	16	1.225	20.5	16.5	36.73469388
6.	Clump 1	13	1.33	39.5	23	18.75
	Clump 2	25.5	1.36	34.5	22.5	29.26829268
7.	Clump 1	18.5	1.35	22.5	16.5	18.18181818
	Clump 2	14	1.1	21.5	16.5	37.5
8.	Clump 1	11.5	1.12	17.5	13	25
	Clump 2	16	1.11	21.5	15.5	38.70967742
9.	Clump 1	12	1.28	30	23	27.27272727
	Clump 2	17.5	0.925	41	24.5	37.5
10.	Clump 1	15.8	1.38	36	23.5	21.95121951
	Clump 2	18	1.57	49	35	18.91891892
11.	Clump 1	3	1.85	23	16.5	3.846153846
	Clump 2	18	0.922	40	25.5	18.42105263
12.	Clump 1	22	1.266	21	15	11.42857143
	Clump 2	18	1.23	29	24	18.91891892



Figure 10 : Root traits of Sugarcane samples



Figure 11 : Root traits of Sugarcane samples

DNA EXTRACTION:

For genomic DNA extraction, fresh leaves were collected from research fields, greenhouses and experimental plots of Indian Institute of Sugarcane Research (ICAR-IISR). In this Experiment, DNA was isolated from fresh leaf of sugarcane tissue using both manual methods And Kit method. The protocol for manual method was followed as described by Murray and Thompson in 1980. In the kit method, the DNA extraction kit by HI Media was used for the extraction of sugarcane DNA.

I. DNA extraction through manual method:

A. Material and reagents required:

- Sugarcane samples
- 2.0 M Tris HCl (pH 8.0)
- 0.5 M EDTA (pH 8.0)
- 10 % CTAB
- 10 % PVP
- 0.2 % β -mercaptoethanol
- Phenol: Chloroform: Isoamyl alcohol (25:24:1)
- Chloroform: Isoamyl alcohol (24:1)
- Isopropanol chilled
- 70 % ethanol chilled
- 1X TE Buffer
- RNase A
- Liquid nitrogen
- Pestle mortar
- Water bath
- Micropipettes
- Eppendorf Tubes
- Centrifuge
- Tube shaker
- Centrifuge tubes

B. Preparation of DNA Extraction Buffer:

Calculation of reagents=

- i. 2M Tris-HCl 50ml

For Tris base, molecular weight= 121.1 g

Therefore, amount of Tris base required to make 2 M solution is

$$\frac{\text{molarity} \times \text{molecular weight} \times \text{volume}}{1000}$$

$$\frac{2 \times 121.11 \times 50}{1000}$$

$$= 12.11 \text{ grams}$$

So, 12.11 grams of Tris-base was dissolved in 25 ml of distilled water. Then pH was Maintained to 8 by adding concentrated HCl and after this, volume was made up to 50 ml.

ii. 10% CTAB (Cetyl dimethyethyl ammonium bromide) 50 ml

In 100 ml, number of CTAB = 10 grams Therefore, in 50 ml, amount of CTAB

$$\frac{50 \times 10}{100} = 5 \text{ grams}$$

So, 5-gram CTAB is dissolved in 50 ml sterile distilled water.

iii. 10% PVP (Polyvinylpyrrolidone) 50 ml

In 100 ml, number of PVP = 10 grams Therefore, in 50 ml, amount of PVP

$$\frac{50 \times 10}{100} = 5 \text{ grams}$$

So, 5 grams of PVP is dissolved in 50 ml of water.

iv. 0.5 M EDTA (ethylene diamine tetra acetic acid di sodium salt)

Molecular weight = 372.2

$$\frac{\text{molarity} \times \text{molecular weight} \times \text{volume}}{1000}$$

$$\frac{0.5 \times 372.2 \times 50}{1000}$$

$$= 9.305 \text{ grams}$$

9.305 grams dissolved in 25 ml of water and then the pH is maintained at 8.0 by NaOH pellets and then make up the volume to 50 ml.

Table 4 : To prepare DNA extraction buffer, volume taken from stock solutions and their final concentration are given in the table

S.No.	Components	Stock concentration	Final concentration	Volume taken from stock solution
1.	Tris HCl (pH 8.0)	2.0 M	200 Mm	5.0 ml
2.	EDTA (pH 8.0)	0.5 M	70 Mm	7.0 ml
3.	CTAB	10% (w/v)	2% (w/v)	10 ml
4.	PVP	10% (v/v)	1% (v/v)	5 ml
5.	β -mercaptoethanol	0.2 (v/v)	0.2% (v/v)	100 μ l
6.	Double distilled water	-	-	22.90 ml
	Total			50 ml

Procedure:

1. Fresh leaf sample was taken for the research field, wiped the leaves with tissue paper to clean the dust and then cut it into fine pieces, folded it in a paper and stored in a deep freezer.
2. Continue the same procedure for the remaining leaf sample and store it in the deep freezer.
3. The cut leaves of 1.0 gram were taken and grinded with liquid nitrogen to fine powder.
4. The ground tissue material was then transferred into a small beaker.
5. Added 5 ml of prewarmed DNA extraction buffer and mixed it properly.
6. Sodium dodecyl sulfate (SDS) of 500 μ l is added and mixed it properly.

10% SDS preparation:

10 gram is dissolved in 100 ml

And for 50 ml = $10 \times 100 \div 50 = 5 \text{ gram}$

5 grams of SDS is dissolved in 50 ml of water.

7. Transferred the mixture to 50 ml centrifuge tube and incubate it at 65°C in water bath for 30-60 minutes with gentle swirling.
8. Equal volume of phenol: chloroform: isoamyl alcohol (25:24:1) is added and mix gently to emulsify
9. for 15-20 minutes.
10. Centrifuged at 8000 rpm in biofuge stators centrifuge machine for 15 min at 10°C.
11. The aqueous phase containing DNA was transferred with a wide bore pipette to an autoclaved new centrifuge tube.
12. Equal volume of chloroform: isoamyl alcohol (24:1) is added and mixed gently to emulsify for 15-20 minutes.

13. Centrifuged at 8000 rpm in biofuge stators centrifuge machine for 15 minutes at 10°C.
14. The aqueous phase containing DNA was transferred with a wide bore pipette to an autoclaved new centrifuge tube.
15. Cold isopropanol of 0.6 ml is added and mixed gently to precipitate the DNA and kept at 0°C overnight.
16. Next day, centrifuged at 8000 rpm for 5 minutes.
17. Gently poured off the upper phase and rinse the DNA pellet in chilled 70% ethanol (1ml) for 20 minutes.
18. Centrifuged again at 8000 rpm for 10 minutes and discarded the solution.
19. Pellet was air dried for 15-20 minutes and dissolved in 250 µl TE buffer.
20. Observe the presence of DNA through gel electrophoresis in 1% agarose gel for 20 minutes at 60 volt and viewed the image through Gel Doc system.

II. **DNA Extraction by HIMEDIA HiGenoMB Plant DNA Isolation Kit (CTAB Method):**

Table 5 : Kit Contents for DNA Extraction by HIMEDIA HiGenoMB Plant DNA Isolation Kit (CTAB Method):

Product Code	Reagents provided	MB502
		50 preps
DS0002	CTAB Extraction Buffer	500 ml
DS0003	RNase A Solution (20 mg/ml)	1.75 ml
DS0004	CTAB Wash Buffer Concentrate	120 ml
DS0040	Elution Buffer(ET) [10Mm Tris-Cl, pH 8.5]	100 ml
DS0029	CTAB Powder	5 grams

Introduction:

DNA extraction from plant tissues, unlike DNA Isolation from mammalian tissues, remains difficult due to the presence of a rigid cell wall surrounding the plant cells. The CTAB method can be used both for freeze-dried leaves and for fresh leaves.

The scale of extraction dependent on the amount of starting material , for e.g., 300-400 mg of material requires 9 ml of Extraction Buffer, which yields up to 500 µg of DNA.

Principle:

CTAB (Cetyl trimethylammonium) , a detergent is used to break open and solubilize the contents of the plant cells. Chlorophyll and some denatured proteins are removed from green plant tissue in an organic chloroform/octanol step, and the organic phase is separated by centrifugation. Since the extract contains DNA and RNA, RNA can be removed by the addition of RNase A, the DNA is precipitated and washed in organic solvents before redissolving in aqueous solution. The concentration of the DNA can be estimated by agarose gel electrophoresis. The DNA recovered is not free from contaminants such as carbohydrates, but is of suitable grade for enzyme digestion, southern blotting and analysis by PCR.

Material Required:

- CTAB Extraction Buffer
- CTAB Wash Buffer
- Elution Buffer
- 2-mercaptoethanol (β -ME)
- Chloroform: isoamyl alcohol (24:1)
- Isopropanol
- Ethanol (96-100 %)
- Ethanol (100 %)

Reagent preparation:**I. CTAB Extraction Buffer:**

90 mg of CTAB Extraction powder is added in 9 ml of CTAB Extraction Buffer and 90 μ l of β -mercaptoethanol is added prior to use. (Temperature to be maintained at 65°C)

For DNA Extraction of 8 samples:

CTAB powder = 180 mg

CTAB Extraction Buffer = 18 ml

β -mercaptoethanol = 180 μ l

II. CTAB Wash Buffer

2 ml of CTAB wash buffer is mixed with 6 ml of ethanol

For DNA Extraction of 8 samples

Stock = 120 ml

1 ml stock needs 360/120 i.e., 3 ml in 1 ml stock

So, 16 ml stock needs

48 ml ethanol to be added in 16 ml stock

III. Chloroform: Isoamyl alcohol (24:1)

3.8 ml of chloroform in 200 μ l of isoamyl alcohol

IV. 70 % ethanol(Add 70 ml of ethanol in 30 ml of distilled water)

Preparation Instructions:

Grinding of the plant material can be done using mortar and pestle with CTAB Extraction Buffer. For plants such as maize and lettuce that have pronounced leaf midribs, these should be removed from the material before grinding, as it is a major source of carbohydrate contamination.

Preparation of CTAB Extraction Buffer:

Immediately prior to use add, 90 μ l of β mercaptoethanol and 90 mg CTAB powder in 9 ml of CTAB Extraction Buffer. Preheat the solution to 65°C.

Chloroform: Octanol (24:1). Store in dark at room temperature (15-25°C). Prepare and dispense this solution in a fume hood.

To the CTAB Wash Buffer concentrate (120 ml) provided, add 360 ml of Ethanol (100%) prior to use. Prechill the CTAB Wash Buffer and 70% ethanol prior to use.

Protocol:

1. To 300-400 mg ground tissue in a sterile centrifuge tube, add 9 ml of prewarmed CTAB Extraction Buffer. Mix gently by inversion.

NOTE: ensure that CTAB powder and β -mercaptoethanol is added to CTAB Extraction Buffer prior to use.

DNA can be extracted from fresh plant tissue by grinding a leaf or leaf disc in a small amount of Extraction Buffer.



Figure 12 : cutting of leaf sample for DNA Extraction



Figure 13 : Grinding of leaf sample for DNA Extraction



Figure 14 : Powdered form of leaf sample for DNA Extraction

2. Incubate the samples for 60-90 minutes, with occasional inversion at 65°C.
3. Allow the samples to cool by keeping it at room temperature (15-25°C) for 5 minutes.
4. Add 5 ml of Chloroform: Isoamyl alcohol (24:1) and mix gently by inverting the tubes for 5 minutes.

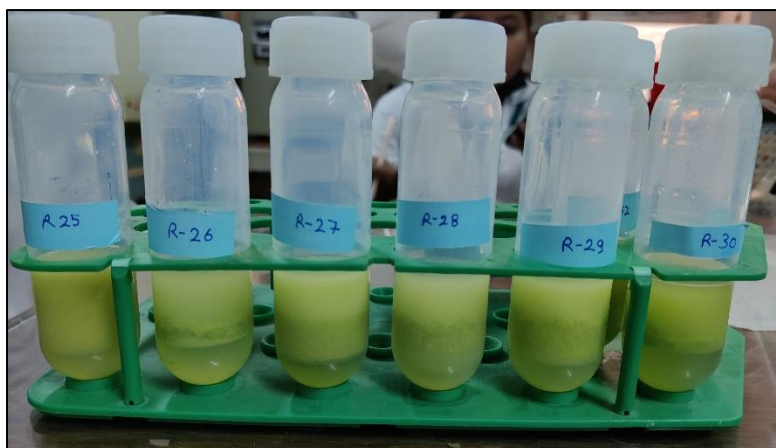


Figure 15 : Addition of Chloroform: Isoamyl alcohol (24:1) to the samples containing CTAB Extraction Buffer



Figure 16 : Addition CTAB wash buffer to the samples containing the DNA pellet

5. Spin the samples at 3000 rpm for 2 minutes at room temperature (15-25°C).
6. Transfer the top aqueous layer into a fresh tube and add 25 µl of RNase A (20 mg/ml). mix the sample gently by inversion and incubate for 30 minutes at room temperature (15-25°C).
7. Add 6 ml of isopropanol to each tube. Mix the samples by inversion until a white fluffy DNA precipitate appears.
8. Centrifuge at 3000 rpm for 5 minutes at room temperature (15-25°C) to pellet the DNA. Discard the supernatant.

9. Resuspend the pellet in 8 ml of cold diluted CTAB wash buffer. Incubate at room temperature (15-25°C) for 20 minutes. Centrifuge at 3000 rpm for 5 minutes at room temperature (15-25°C). discard the supernatant
10. Add 8 ml of cold 70 % ethanol to the tube containing DNA. Centrifuge at 3000 rpm for 5 minutes at room temperature (15-25°C). Discard the supernatant.
11. Air dry the pellet to remove the traces of ethanol for about 10 minutes.
12. Transfer the DNA sample to a tube containing 1 ml Elution Buffer (ET). Dissolve the pellet gently by pipetting.



Figure 17 : Pellet

INTRODUCTION TO AGAROSE GEL ELECTROPHORESIS

Agarose gel electrophoresis is a method of gel electrophoresis used in biochemistry, molecular biology and clinical chemistry to separate a mixed population of DNA or proteins in a matrix of agarose. The proteins may be separated by charge and/or size (isoelectric focusing agarose electrophoresis is essentially size independent), and the DNA and RNA fragments by length. Biomolecules are separated by applying an electric field to move the charged molecules through an agarose matrix, and the biomolecules are separated by size in the agarose gel matrix.

Material required:

- Electrophoresis unit
- Cast tray
- 2% agarose gel (1.6 gm agarose in 80 ml 1 X TAE Buffer)
- Ethidium bromide (EtBr)
- 1 X TAE Buffer
- Micropipette
- Tips
- DNA samples
- Ladder (50 bp and 100 bp)

Procedure to prepare stock solution of TAE buffer:

Make a concentrated (50X) stock solution of TAE by weighing out 242-gram Tris base (Molar Mass = 121.14) and dissolving in approximately 750 ml deionized water. Carefully add 57.1 ml glacial acid and add 100 ml of 0.5 M EDTA (pH 8.0) and adjust the solution to a final volume of 1 L. This stock solution can be stored at room temperature. The pH of this buffer is not adjusted and should be about 8.5

Gel preparation:

- For preparation of 2% agarose gel , 1 gram agarose was weighed and dissolved in 50 ml TAE buffer
- The gel solution was boiled in the microwave oven until it became clear and transparent.
- 1 µl ethidium bromide was added to the molten agarose wearing gloves and mixed gently without forming bubbles
- Molten agarose was cooled before pouring it in the agarose gel slab.

Casting agarose gel:

- Gel caster was leveled on a flat surface.

- The movable wall was disengaged and the open end of the gel caster was slid by turning and lifting the cam peg upward.
- The open edge of the tray was placed against the fixed wall of the gel caster.
- The movable wall was slid against the edge of the tray.
- To seal the open tray ends, the cam peg was engaged by turning and pressing downward simultaneously.
- When the cam peg was dropped into the appropriate slot, the peg was turned in either direction until resistance was felt; this action seals the edges of the tray for casting.
- The comb was placed into the appropriate slot of the tray.
- The prepared gel was poured into the tray and left to solidify.
- After solidification, the cam peg was loosened to remove the tray from the caster.
- The gel tray was dipped into the electrophoresis unit which was pre-filled with TAE buffer.

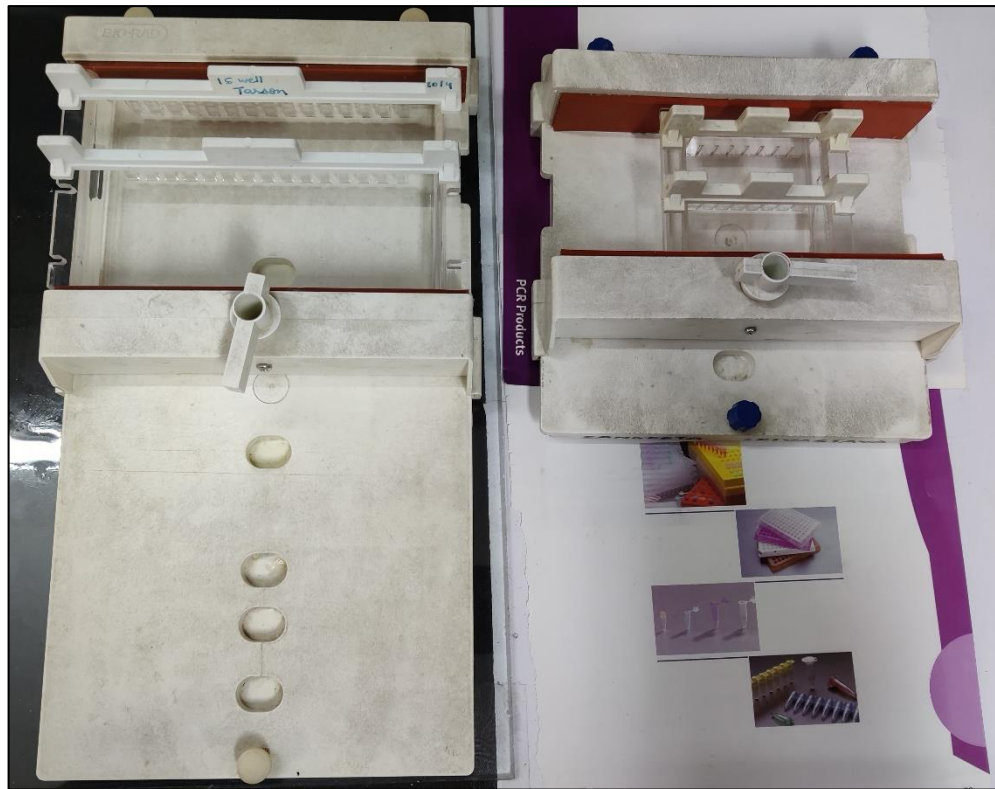


Figure 32 : Casting agarose gel

Loading the sample:

Sample was loaded along with a tracking dye usually bromophenol blue or xylene cyanol. 6X loading dye solution was used consisting of both the dyes. The sample was well mixed with the dye before loading on the well of the electrophoresis unit.

Setting up and running the Electrophoresis Unit:

After the agarose gel solidifies, sample loading and electrophoresis can be done. Agarose gels can be run in many different types of electrophoresis buffers. Nucleic acid agarose gel

electrophoresis is usually conducted with either Tris-Acetate-EDTA (TAE) buffer or Tris Borate- EDTA (TBE) buffer provide faster electrophoretic migration of linear DNA and better resolution of super coiled DNA. TBE buffer have a stronger buffering capacity for longer or higher voltage electrophoresis runs.

Steps involved:

- Samples for gel loading were prepared. Loading volume is dependent upon the type of comb used (i.e., well thickness and length) and thickness of the gel.
- Standard nucleic acid sample loading dye was added to make the sample dense for under laying into sample wells
- The samples were loaded into the wells using standard micropipettes
- The lid was placed on the unit carefully.
- The gel was run for 20 minutes at 60 volts.
- The run was stopped before the bromophenol blue loading dye front exits the gel.
- The gel was visualized under U.V. light

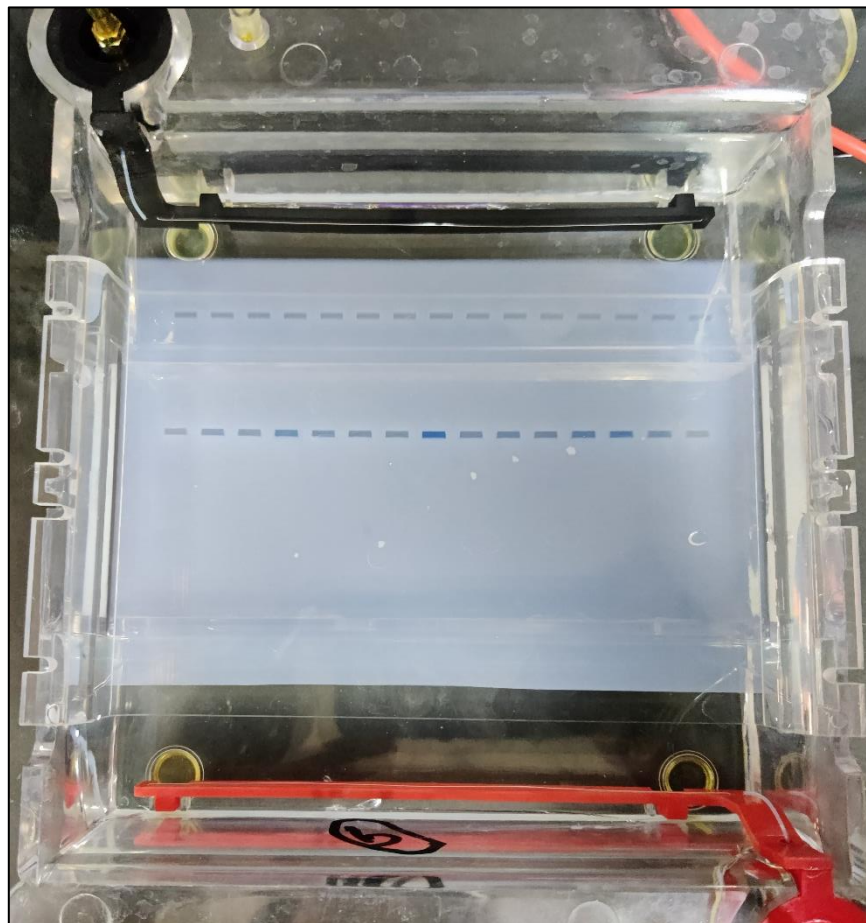


Figure 33 : Samples loaded in the wells on the agarose gel

GEL DOCUMENTATION SYSTEM

Gel Doc, also known as gel documentation system, gel image system or gel imager, is widely used in molecular biology laboratories for the imaging and documentation of nucleic acid and protein polyacrylamide or agarose gels typically stained with ethidium bromide or other fluorophores. Generally, a gel doc is composed of an ultraviolet (UV) light transilluminator, a hood to shield external light sources and a camera for image capturing.

Steps involved:

- The gel was placed inside gel documentation system, for the imaging and documentation of agarose gel typically stained with ethidium bromide.
- The ethidium bromide fluoresced reddish-orange in the presence of DNA, since it has intercalated within the DNA.
- The gel was then photographed and saved for further analysis.



Figure 34 :

Agarose gel with DNA samples placed inside the gel documentation system, gel image system or gel imager after electrophoresis

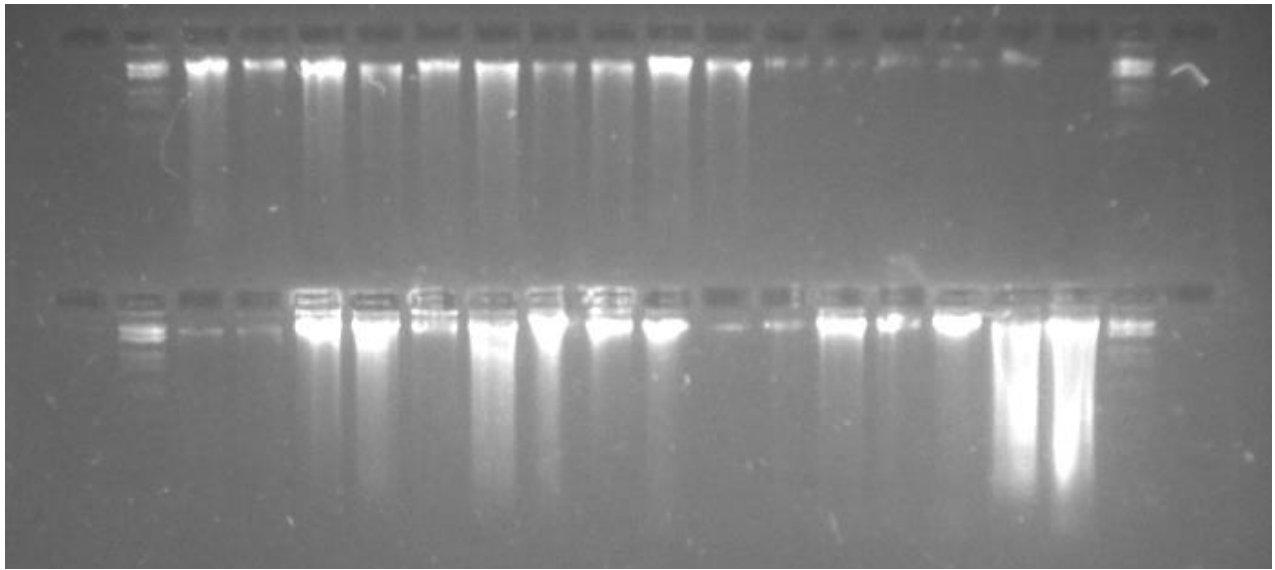


Figure 35 :
Sugarcane DNA samples
Left and Right- λ DNA ladder

NANODROP SPECTROPHOTOMETER

Principle:

It operates on Beer's Law. The sample retention system Nanodrop Spectrophotometry employs surface tension to hold the sample in place between two optical fibers. This enables the measurement of very highly concentrated samples without the need for dilutions.

The Nanodrop Spectrophotometer works using sample combining fiber optic technology and natural surface tension properties which capture and retain small amounts of sample that does not require any apparatus such as cuvettes or capillaries. Therefore, the system provides shorter path lengths, which result in a broad range of nucleic acid concentration measurements, essentially eliminating the need to perform dilutions. Due to the reduction in amount of sample that is required for analysis facilitates inclusion of additional quality control steps throughout many molecular workflows and also increase the efficiency and ultimately leading to greater confidence in downstream results.



Figure 18 : Nanodrop Spectrophotometer

Nanodrop micro volume technology employs a sample retention system that relies on the surface tension properties of the sample being measured to form a liquid column. It is essential that the sample makes contact with the upper and lower optical measurement surfaces for proper column formation. If at any point the measurement results seem inaccurate or not reproducible, it is most likely the result of sample heterogeneity or liquid column breakage.

Concentration and Purity:

Ratio is dependent on the pH and ionic strength of the buffer used to make the blank and sample measurements. Purity is measured under the 260/280 column

Protocol:

- 1) 1 μ l of clean deionized water was pipetted and the upper and lower optical surfaces were wiped.
- 2) While closing the lever arm, it was checked whether the upper pedestal is in contact with the deionized water and using dry, lint free lab wipe, both the optical surfaces were cleaned after lifting the lever arm.

- 3) Open the software and go to Nucleic Acid application. A small-volume, calibrated pipette was used to perform a blank test by dispensing 1 μL of buffer onto the lower optical surface. The lever arm was lowered and "Blank" in the Nucleic Acid application was selected.
- 4) As soon as the blank measurement completes, both the optical surface was wiped and cleaned.

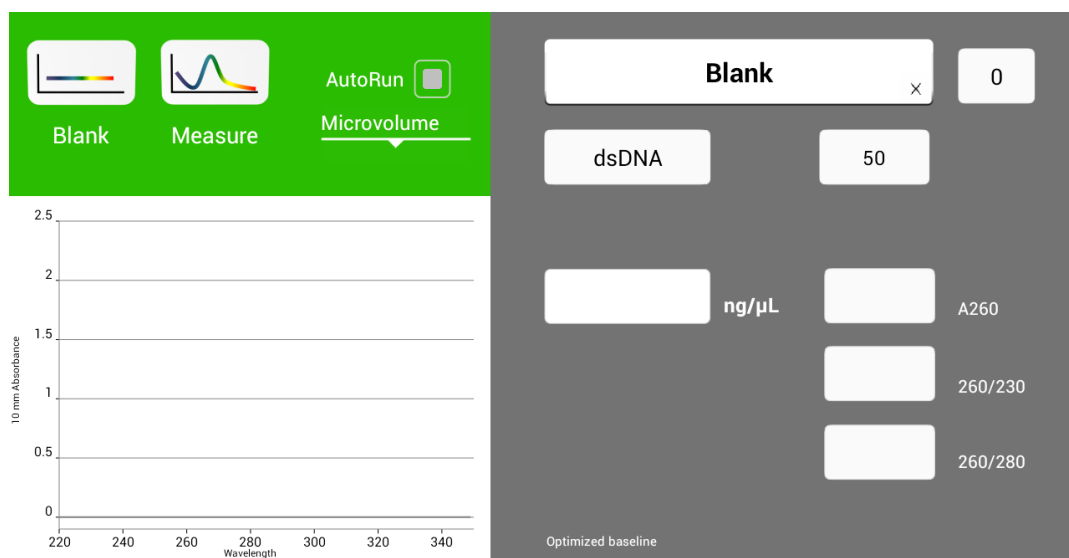


Figure 18 : Measurement of Blank on Nanodrop Spectrophotometer

- 5) The appropriate constant for the sample that is to be measured was chosen.
- 6) 1 μL of nucleic acid sample onto the lower optical pedestal was dispensed and closed the lever arm. Because the measurement is volume independent, the sample only needs to bridge the gap between the two optical surfaces for a measurement to be made.
- 7) Select "Measure" in the application software. The software will automatically calculate the nucleic acid concentration and purity ratios. Following sample measurement, review the spectral output.
- 8) The software will automatically calculate the nucleic acid concentration and purity ratios. Following sample measurement, review the spectral image to assess sample quality.
- 9) A typical nucleic acid sample will have a very characteristic profile.
- 10) To accurately assess sample quality, 260/280 or 260/230 ratios should be analyzed in combination with overall spectral quality. Pure nucleic acids typically yield a 260/280 ratio of ~ 1.8 and a 260/280 ratio of ~ 2.0 for DNA and RNA, respectively. This ratio is dependent on the pH and ionic strength of the buffer used to make the blank and sample measurements. Acidic solutions will under-represent the ratio by 0.2-0.3, while a basic solution will over-represent the ratio by 0.2-0.3. The 260/230 purity ratio is a second measure of DNA purity with values for a "pure" nucleic acid commonly in the range of 1.8-2.2. Purity ratios that are significantly lower than the expected values may indicate the isolation technique used may require further optimization

Table 6 : Measurement of concentration of 96 sugarcane varieties on Nanodrop Spectrophotometer

S.no	Variety name	Concentration (ng/μl)
1.	Variety 01	740.47
2.	Variety 02	580.852
3.	* Variety 03	155.026
4.	Variety 04	652.326
5.	* Variety 05	60.122
6.	Variety 06	523.406
7.	Variety 07	552.614
8.	Variety 08	392.524
9.	Variety 09	715.67
10.	Variety 10	368.41
11.	Variety 11	534.178
12.	Variety 12	406.212
13.	Variety 13	424.206
14.	Variety 14	406.285
15.	Variety 15	557.696
16.	* Variety 16	177.234
17.	* Variety 17	175.47
18.	Variety 18	346.886
19.	* Variety 19	65.934
20.	Variety 20	344.638
21.	Variety 21	418.897
22.	Variety 22	628.953
23.	Variety 23	377.352
24.	Variety 24	221.228
25.	* Variety 25	185.163
26.	* Variety 26	146.808
27.	* Variety 27	99.418
28.	Variety 28	675.122
29.	Variety 29	478.674
30.	Variety 30	604.424
31.	* Variety 31	2.408
32.	* Variety 32	21.151
33.	Variety 33	562.617
34.	Variety 34	492.079
35.	Variety 35	465.327
36.	* Variety 36	178.746
37.	Variety 37	276.394
38.	Variety 38	416.602
39.	Variety 39	561.429
40.	Variety 40	344.269
41.	Variety 41	534.812
42.	* Variety 42	45.691
43.	* Variety 43	196.065
44.	Variety 44	310.011
45.	Variety 45	313.745
46.	Variety 46	305.402

47.	Variety 47	301.898
48.	Variety 48	262.88
49.	* Variety 49	143.866
50.	* Variety 50	86.01
51.	* Variety 51	71.428
52.	Variety 52	952.171
53.	* Variety 53	62.519
54.	Variety 54	646.265
55.	Variety 55	1417.485
56.	* Variety 56	169.331
57.	Variety 57	929.818
58.	* Variety 58	193.819
59.	Variety 59	401.11
60.	Variety 60	610.64
61.	Variety 61	438.192
62.	Variety 62	443.776
63.	Variety 63	1093.943
64.	Variety 64	2742.284
65.	Variety 65	368.493
66.	Variety 66	301.012
67.	Variety 67	482.087
68.	Variety 68	481.77
69.	* Variety 69	105.921
70.	* Variety 70	34.131
71.	* Variety 71	172.761
72.	Variety 72	393.404
73.	Variety 73	597.108
74.	Variety 74	740.904
75.	Variety 75	498.176
76.	Variety 76	528.475
77.	Variety 77	719.291
78.	Variety 78	539.103
79.	Variety 79	672.567
80.	Variety 80	290.427
81.	Variety 81	310.185
82.	Variety 82	1043.461
83.	Variety 83	939.841
84.	Variety 84	490.36
85.	Variety 85	1493.868
86.	Variety 86	497.703
87.	Variety 87	1644.207
88.	Variety 88	775.845
89.	Variety 89	795.165
90.	Variety 90	790.916
91.	Variety 91	527.085
92.	Variety 92	302.634
93.	Variety 93	1049.701
94.	Variety 94	656.495
95.	* Variety 95	168.573
96.	Variety 96	238.89

*these varieties have poor concentrations



Figure 19 : Graph of Varieties 1-9 on Nanodrop Spectrophotometer



Figure 20 : Graph of Varieties 10-18 on Nanodrop Spectrophotometer



Figure 21 : Graph of Varieties 19-27 on Nanodrop Spectrophotometer



Figure 22 : Graph of Varieties 28-36 on Nanodrop Spectrophotometer



Figure 23 : Graph of Varieties 37-45 on Nanodrop Spectrophotometer



Figure 24 : Graph of Varieties 46-54 on Nanodrop Spectrophotometer



Figure 25 : Graph of Varieties 55-63 on Nanodrop Spectrophotometer



Figure 26 : Graph of Varieties 64-72 on Nanodrop Spectrophotometer



Figure 27 : Graph of Varieties 73-81 on Nanodrop Spectrophotometer



Figure 28 : Graph of Varieties 82-90 on Nanodrop Spectrophotometer



Figure 29 : Graph of Varieties 91-96 on Nanodrop Spectrophotometer

Table 7 : Concentration of 5 sample varieties

S.No.	Variety Name	Sample	Concentration (ng/μl)
1.	Variety 32	R-33	359.668
2.	Variety 33	R-34	251.820
3.	Variety 34	R-35	331.181
4.	Variety 35	R-36	165.491
5.	Variety 36	R-37	298.159

#	Sample Name	Conc.	Units
BLANK			
1		94.341	ng/μL
2	R-33	359.668	ng/μL
3	R-34	251.820	ng/μL
4	R-35	331.181	ng/μL
7	R-36	165.491	ng/μL
8	R-36	298.159	ng/μL

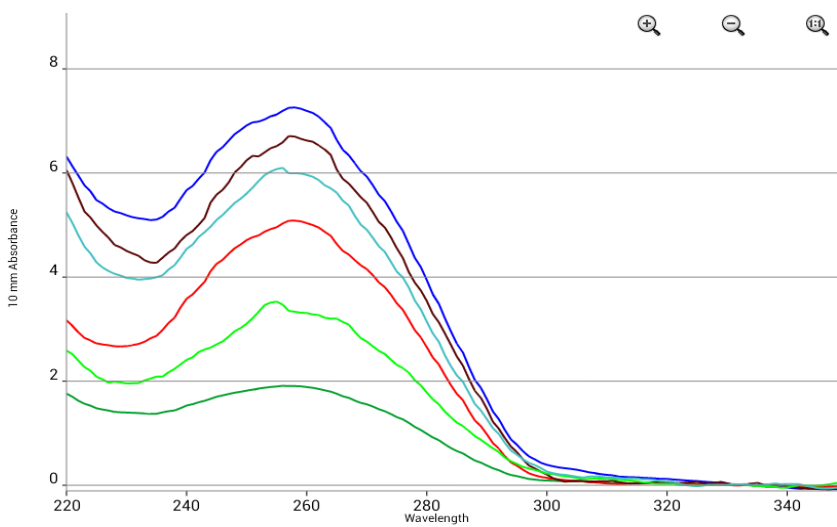


Figure 30 : Combined Graph of Varieties 33-37 on Nanodrop Spectrophotometer

Calculation:

Formula to be used,

$$N_1 V_1 = N_2 V_2$$

- R-33
 $N_1 = 50 \text{ ng/}\mu\text{l}$
 $V_1 = 100 \text{ }\mu\text{l}$
 $N_2 = 359.668 \text{ ng/}\mu\text{l}$

$$V_2 = \frac{50 \times 100}{359.668} = 13.90 \text{ }\mu\text{l}$$

- R-34
 $N_1 = 50 \text{ ng/}\mu\text{l}$
 $V_1 = 100 \text{ }\mu\text{l}$
 $N_2 = 251.820 \text{ ng/}\mu\text{l}$

$$V_2 = \frac{50 \times 100}{251.820} = 19.85 \text{ }\mu\text{l}$$

- R-35
 $N_1 = 50 \text{ ng/}\mu\text{l}$
 $V_1 = 100 \text{ }\mu\text{l}$
 $N_2 = 331.181 \text{ ng/}\mu\text{l}$

$$V_2 = \frac{50 \times 100}{331.181} = 15.09 \text{ }\mu\text{l}$$

- R-36
 $N_1 = 50 \text{ ng/}\mu\text{l}$
 $V_1 = 100 \text{ }\mu\text{l}$
 $N_2 = 165.491 \text{ ng/}\mu\text{l}$

$$V_2 = \frac{50 \times 100}{165.491} = 30.21 \text{ }\mu\text{l}$$

- R-37
 $N_1 = 50 \text{ ng/}\mu\text{l}$
 $V_1 = 100 \text{ }\mu\text{l}$
 $N_2 = 298.159 \text{ ng/}\mu\text{l}$

$$V_2 = \frac{50 \times 100}{298.159} = 16.76 \text{ }\mu\text{l}$$

ALIUQUOT FORMATION

Dilution of DNA: To prepare 100 μl of diluted DNA

Table 8 : Aliquot formation of 5 sample varieties by Dilution of DNA

S no.	Variety name	Sample	Double Distilled Water (μl)	DNA (μl)
1.	Variety 33	R-33	86.1	13.90
2.	Variety 34	R-34	80.15	19.85
3.	Variety 35	R-35	84.91	15.09
4.	Variety 36	R-36	69.79	30.21
5.	Variety 37	R-37	83.24	16.76



Figure 31 : Dilution of DNA

POLYMERASE CHAIN REACTION (PCR)

The polymerase chain reaction (PCR) is a scientific technique in molecular biology to Amplify a single or a few copies of a piece of DNA across several orders of magnitude, Generating thousands to millions of copies of a particular DNA sequence. Polymerase Chain Reaction was developed in 1984 by the American biochemist Kary Mullis. Progress was limited by primer synthesis and polymerase purification issues. PCR is now a common and often indispensable technique used in medical and biological research labs for a variety of applications. The polymerase chain reaction is a powerful technique that has rapidly become one of the most widely used techniques in molecular biology because it is quick, inexpensive and Simple. The technique amplifies specific DNA fragments from minute quantities of source DNA Material, even when that source DNA is of relatively poor quality.

PCR is a quick and easy method for generating unlimited copies of any fragment of DNA. From the daily practicalities of medical diagnosis to the theoretical framework of systematics, from courts of law to field studies of animal behavior, PCR takes analysis of tiny amounts of genetic material - even damaged genetic material to a new level of precision and reliability.

Principle:

The PCR technique is dependent on the enzymatic replication of DNA. In PCR, a short segment of DNA is amplified using primer mediated enzymes. DNA polymerase synthesizes new strands of DNA complementary to the template DNA. The DNA polymerase can add a nucleotide to the pre-existing 3'-OH group only. Therefore, a primer is required. Thus, more nucleotides are added to the 3' prime end of the DNA polymerase.

Components of PCR

- **DNA Template–**

the DNA of interest from the sample.

- **DNA Polymerase enzyme –**

Makes new strands of DNA, using existing strands as templates. Catalyzes synthesis of DNA and helps in proof-reading. The DNA polymerase attaches at the end of the primer and adds nucleotides, brings in nucleotides and puts them in the correct order. Taq Polymerase is used as it can work on high temperatures with high efficiency and amplification capacity.

- **Oligonucleotide Primers-**

These are the short stretches of single-stranded DNA complementary to the 3' ends of sense and anti-sense strands. A pair of primers is used to hybridize with the sample DNA and define the region of the DNA that will be amplified. Two primers are used in each PCR reaction, and they are designed so that they flank the target region (region that should be

copied). They are given sequences that will make them bind to the opposite strands of the template DNA, just as the edges of the region to be copied. The forward primer attaches to the start codon of the template DNA (anti-sense strand). The reverse primer attaches to the stop codon of the complementary strand of the DNA (the sense strand)

- **Deoxyribonucleotide triphosphate–**

These provide energy for polymerization and are the building blocks for the synthesis of DNA. These are single units of bases. Provide nucleotides to the unzipped strand using the template of a single side. This turns a single strand of DNA into two, and can continue exponentially as long as reagents remain present until the final hold stage.

- **Buffer System–**

Magnesium and Potassium provide optimum conditions for DNA denaturation and renaturation. It is also important for fidelity, polymerase activity, and stability.

- **MgCl₂ -**

MgCl₂ is an essential cofactor that enhances the activity of Taq DNA polymerase which in turn increases the amplification rate of DNA.

PCR Steps:

The PCR involves three major cyclic reactions:

Denaturation

Denaturation occurs when the reaction mixture is heated to 94°C for about 0.5 to 2 minutes. This breaks the hydrogen bonds between the two strands of DNA and converts it into a single-stranded DNA. The single strands now act as a template for the production of new strands of DNA. The temperature should be provided for a longer time to ensure the separation of the two strands.

Annealing

The reaction temperature is lowered to 54-60°C for around 20-40 seconds. Here, the primers bind to their complementary sequences on the template DNA. Primers are single-strand sequences of DNA or RNA around 20 to 30 bases in length. They serve as the starting point for the synthesis of DNA. The two separated strands run in the opposite direction and consequently there are two primers- a forward primer and a reverse primer.

Elongation

At this step, the temperature is raised to 72-80°C. The bases are added to the 3' end of the primer by the Taq polymerase enzyme. This elongates the DNA in the 5' to 3' direction. The DNA polymerase adds about 1000bp/minute under optimum conditions. Taq Polymerase can tolerate very high temperatures. It attaches to the primer and adds DNA bases to the single strand. As a result, a double-stranded DNA molecule is obtained. These three steps are repeated 20-40 times in order to obtain a number of sequences of DNA of interest in a very short time period

Master Mix Preparation for PCR:

Table 9 : Master Mix Preparation for PCR

S. No.	Components	Volume per sample (in μL) * No. of samples
1.	Template DNA	1
2.	Double Distilled Water	0.9×1
3	PCR Buffer	1.0×1
4.	dNTP	0.8×1
5.	MgCl ₂	1.0×1
6.	Primer: • Forward primer • Reverse primer	0.4×1 0.4×1
7.	Taq Polymerase	4.5×1
Total		10 μL

PCR Procedure

1. Master Mix was prepared for each primer combination accordingly as mentioned above and it was vortexed for proper mixing of the contents.
2. PCR tube stand was taken and 12 PCR tubes were arranged in each of the stand and labelled from 1-12 according on all the stands.
3. 1 μL of DNA sample was dispensed in the respective PCR tubes as labelled.
4. 9 μL of master mix was dispensed in each 12 PCR tubes respectively.
5. Then all the PCR tubes of PCR stand was spined quickly for 15 seconds approx. to mix the master mix with the DNA evenly.
6. The PCR tubes were arranged in the PCR tray.
7. The tray was kept in the Gradient Cycle for the given time.
8. The suitable program was set.
9. 4 μL of PCR product (1 μL loading + 3 μL PCR product) was loaded in 2% Agarose gel and run at 60 Volt for 40 minutes.
10. The result was observed via Gel Doc System.

11. Then the same procedure was followed with the other set of DNA sample containing different primer combinations and the result was observed by loading it in 2% Agarose gel.
12. The result was saved for further studies.

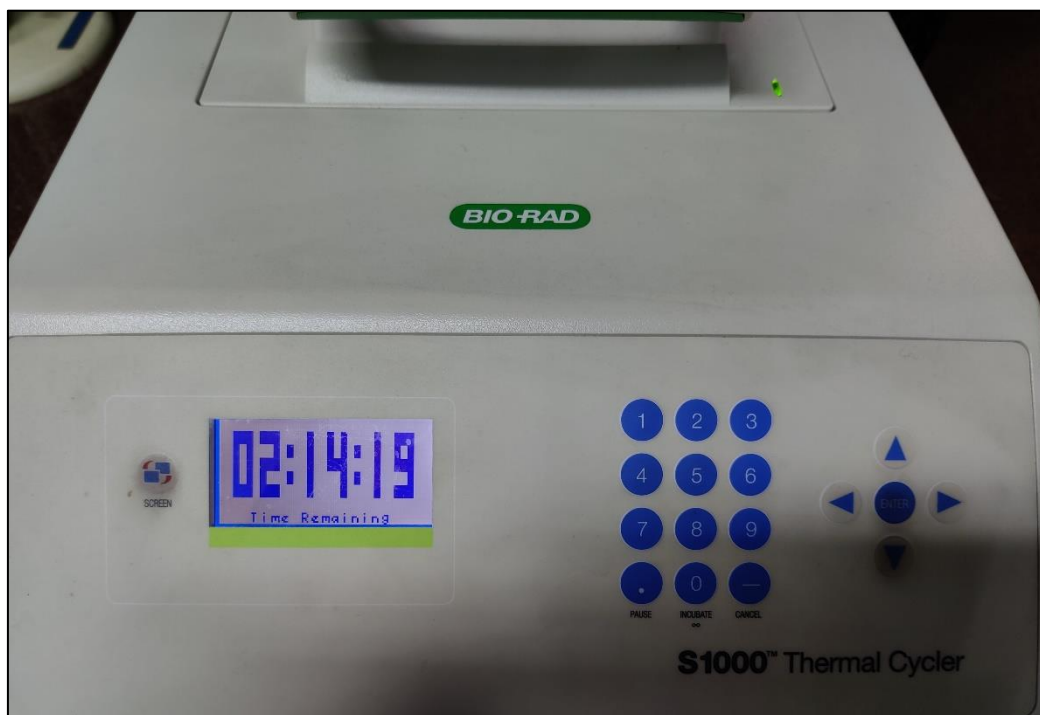


Figure 36 : Thermo Cycler

PCR program:

The thermo cycler was set to 8 cycles of denaturation at 94°C for 45 sec, annealing temperature 35°C for 45 seconds, extension at 72°C for 1 minute, followed by 35 cycles of denaturation at 94°C for 45 seconds, annealing temperature 52°C for 45 seconds, extension at 72°C for 1 minute. The run included an initial denaturation at 94°C for 4 minutes and final extension at 72°C for 7 minutes.

Table 10 : PCR Program

STEP	TEMPERATURE (°C)	NO. OF CYCLES	DURATION
Initial Denaturation	94	1	4 minutes
Denaturation	94	8	45 seconds
Annealing	35		45 seconds
Extension	72		1 minutes
Denaturation	94	35	45 seconds
Annealing	52		45 seconds
Extension	72		1 minute
Final Extension	72	1	7 minutes
Hold	4		10 minutes

RESULT:

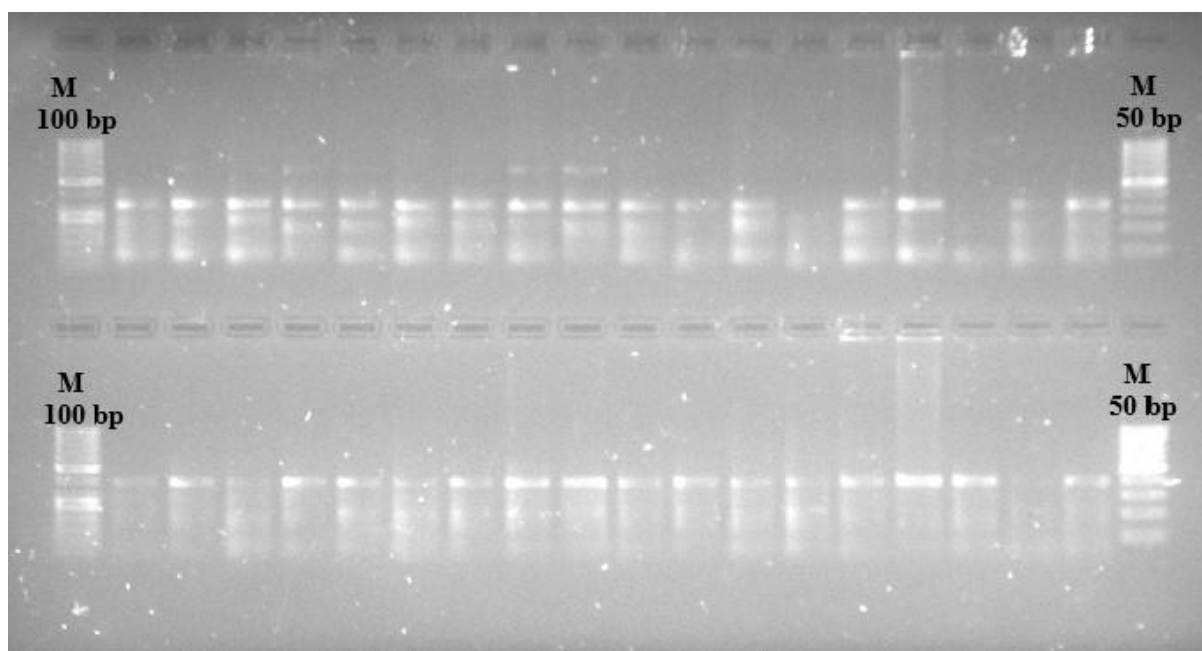


Figure 37 :

***PCR of 18 samples with primer pair 1 (upper gel) and primer pair 2 (Lower gel),
Left-100 bp ladder, Right-50 bp ladder***